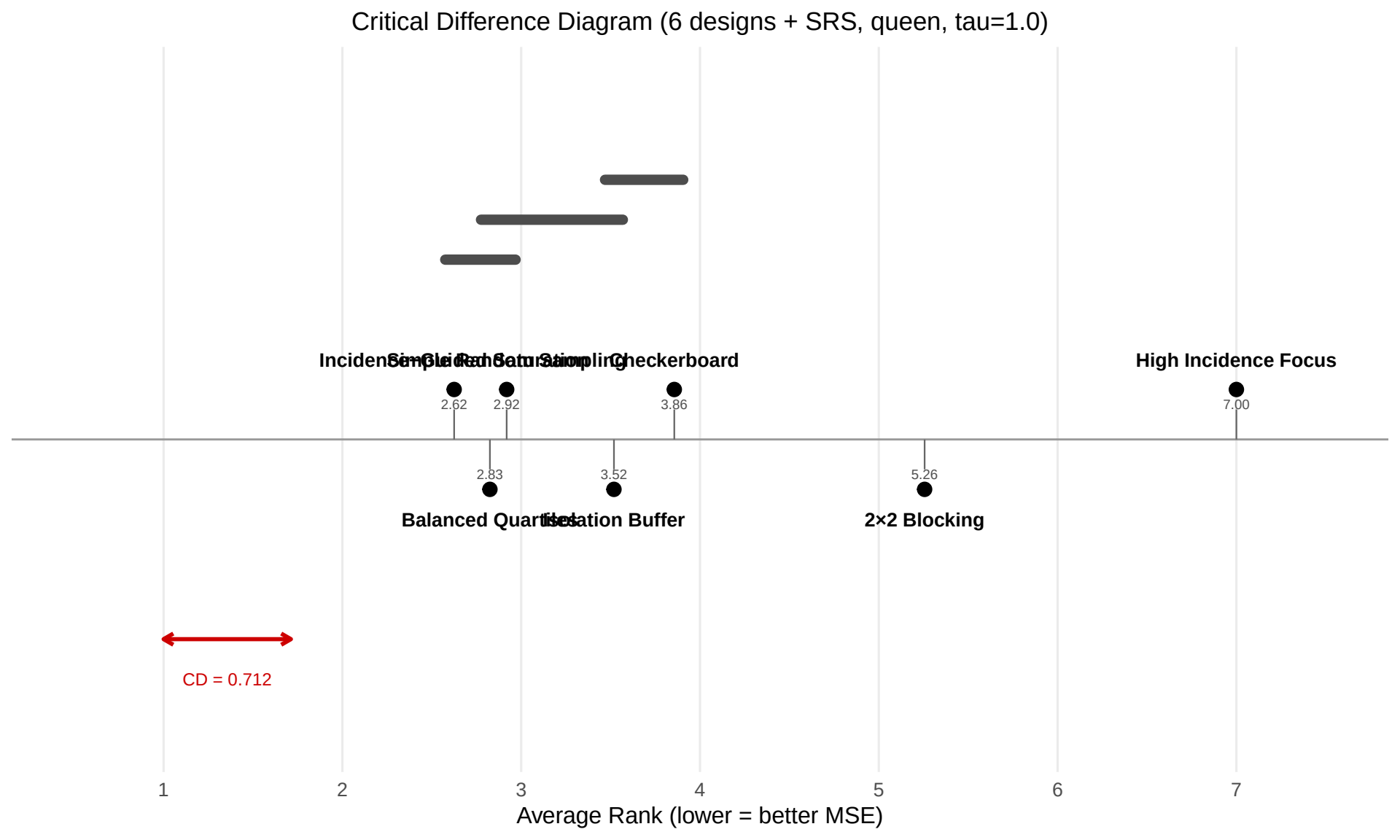
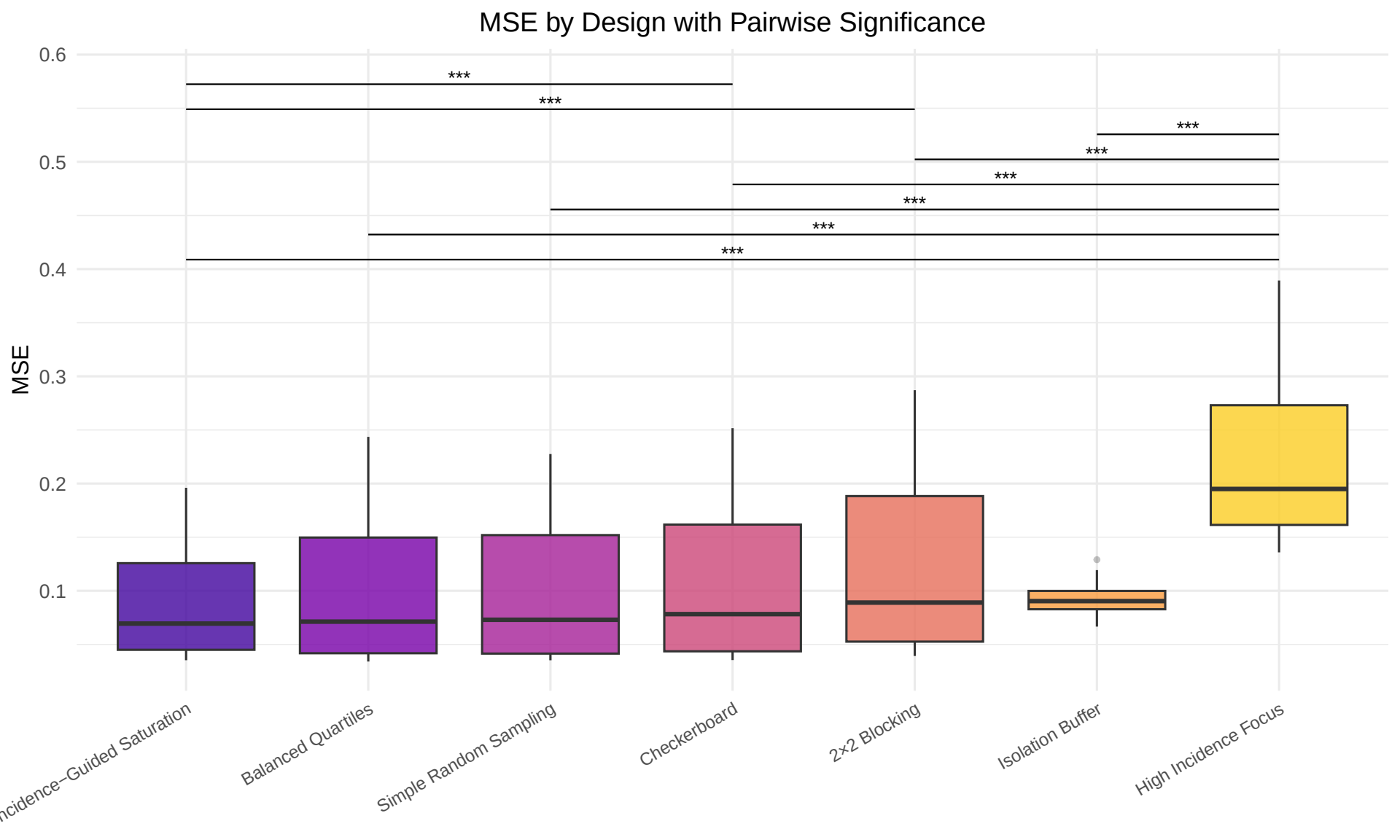
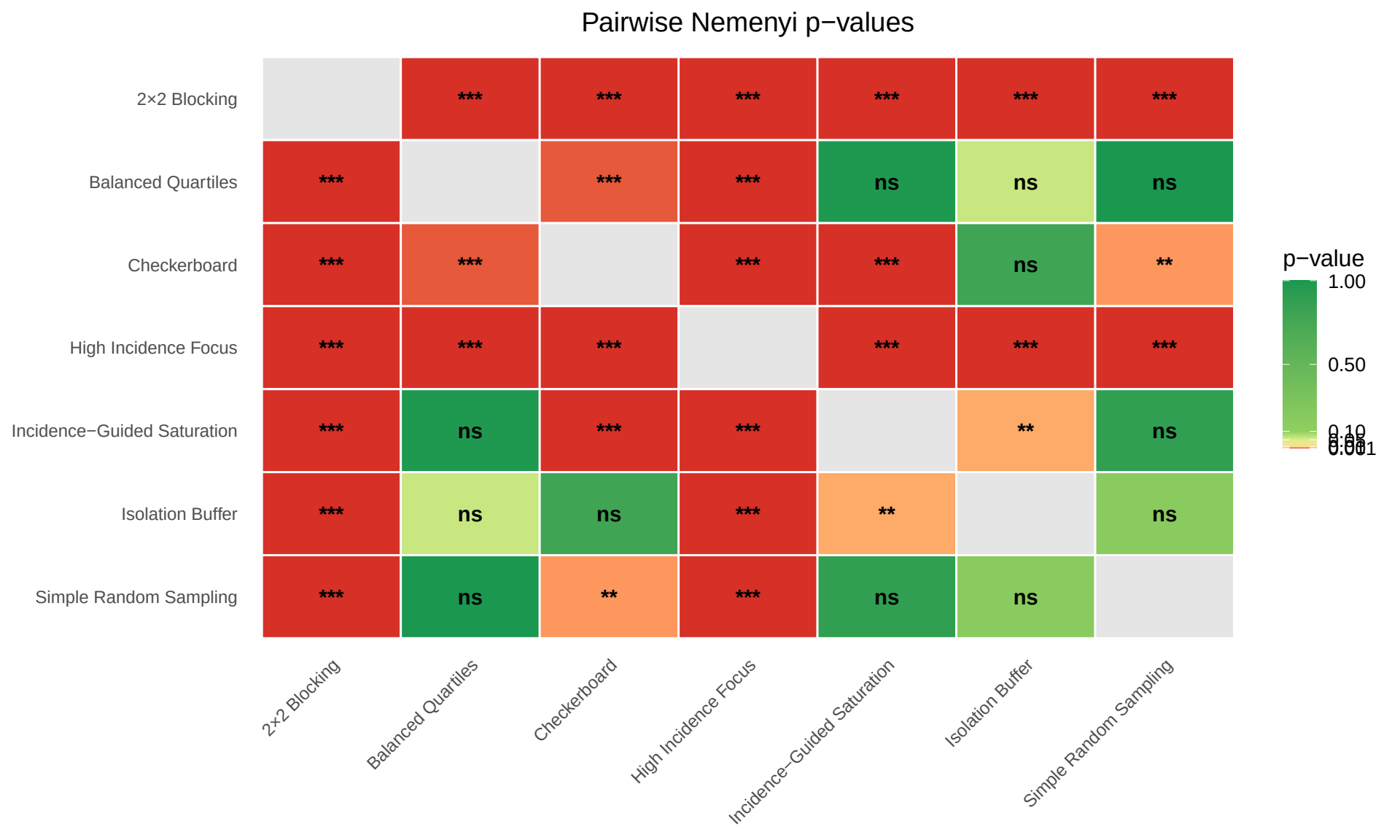
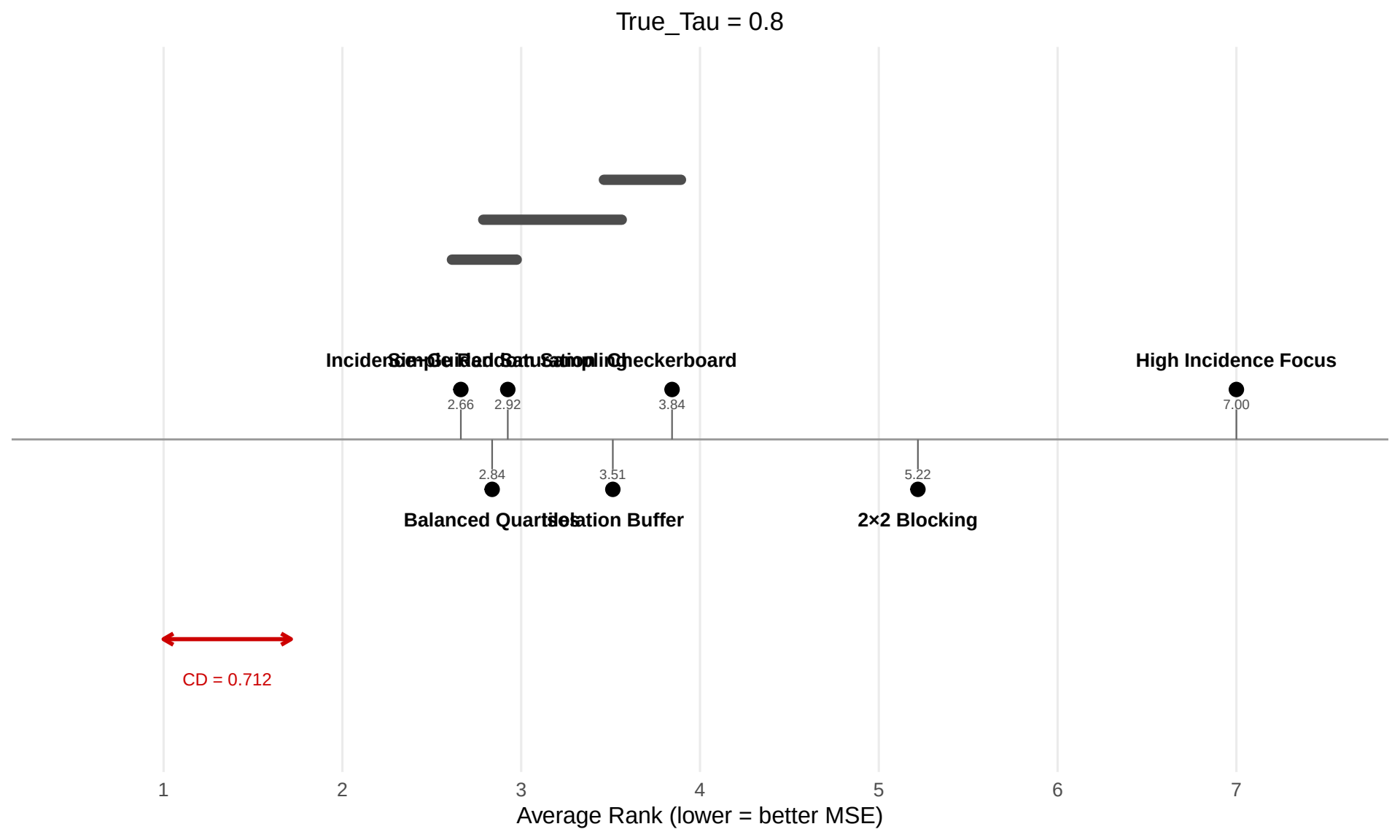
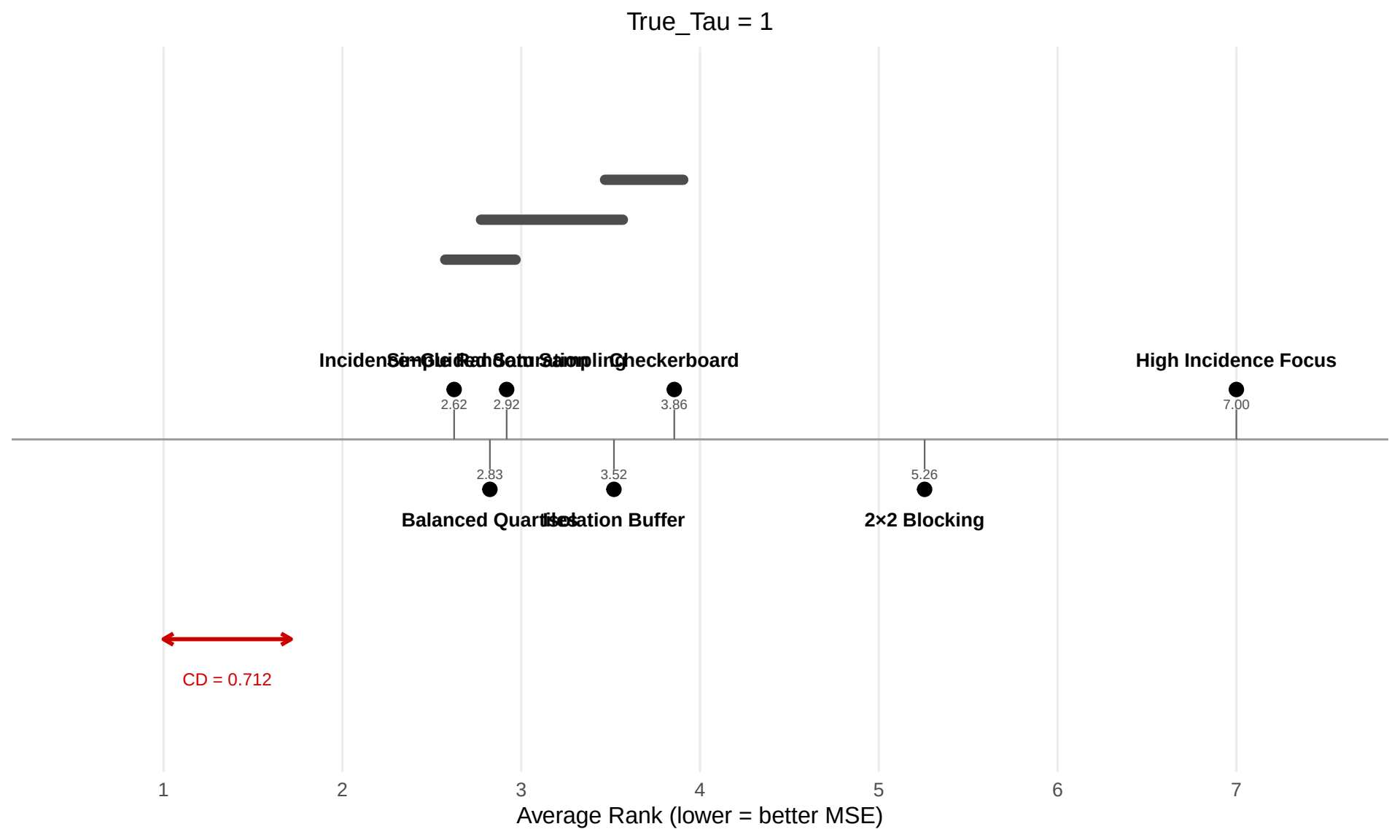

Critical Difference Diagram (6 designs + SRS, queen, tau=1.0)
Checkerboard
High Incidence Focus
2.62
2.92
3.86
7.00
2.83
3.52
5.26
Balanced Quartiles
Isolation Buffer
2×2 Blocking
CD = 0.712
1
2
3
4
5
6
7
Average Rank (lower = better MSE)

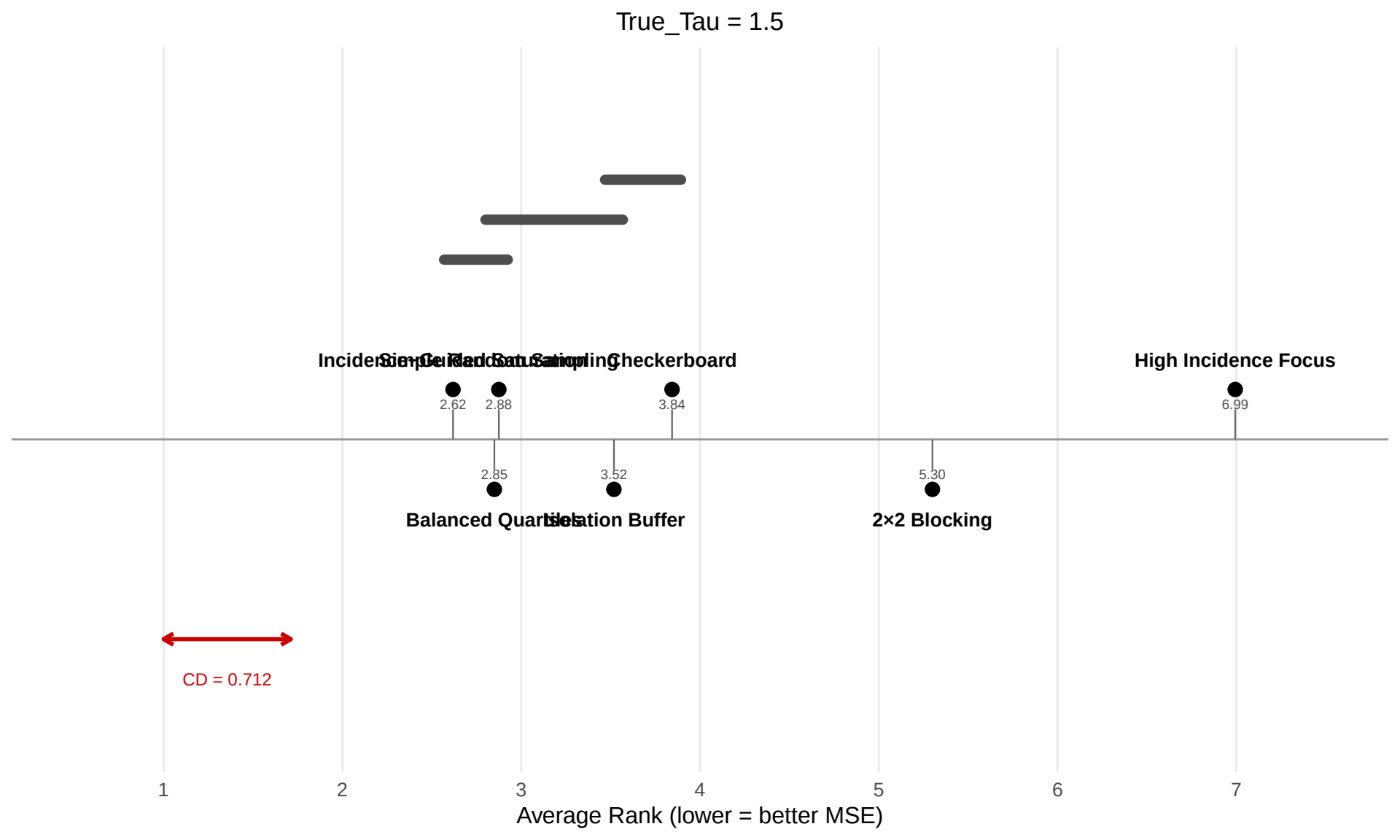

MSE by Design with Pairwise Significance
MSE
0.6
0.5
0.4
0.3
0.2
0.1
***
***
***
***
***
***
***
ncidence−Guided Saturation
Balanced Quartiles
Simple Random Sampling
Checkerboard
2×2 Blocking
Isolation Buffer
High Incidence Focus

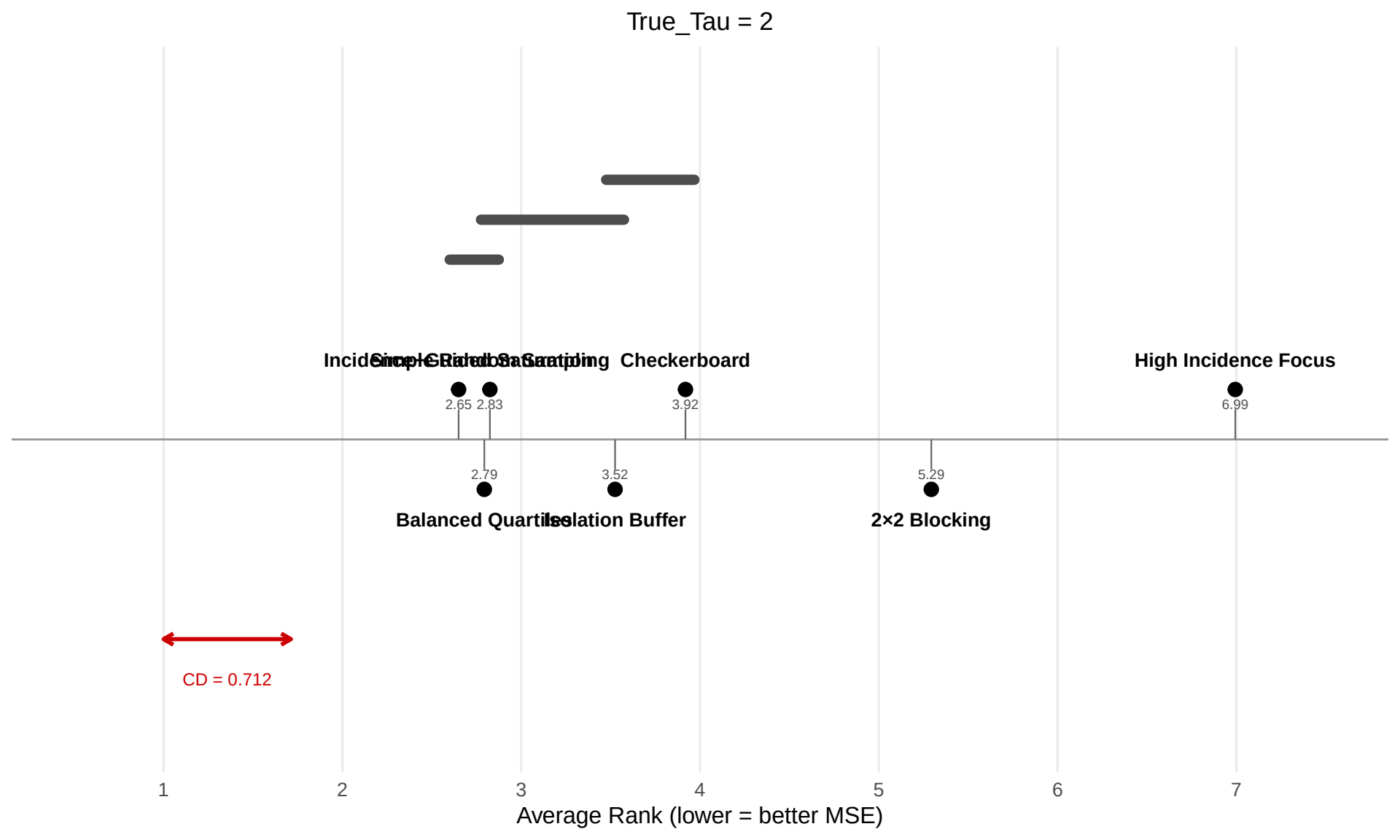
Pairwise Wilcoxon (Holm) p−values (Holm−adjusted)
2×2 Blocking
Balanced Quartiles
Checkerboard
High Incidence Focus
Incidence−Guided Saturation
Isolation Buffer
Simple Random Sampling
***
ns
*
p−value
1.00
0.50
0.10
0.001

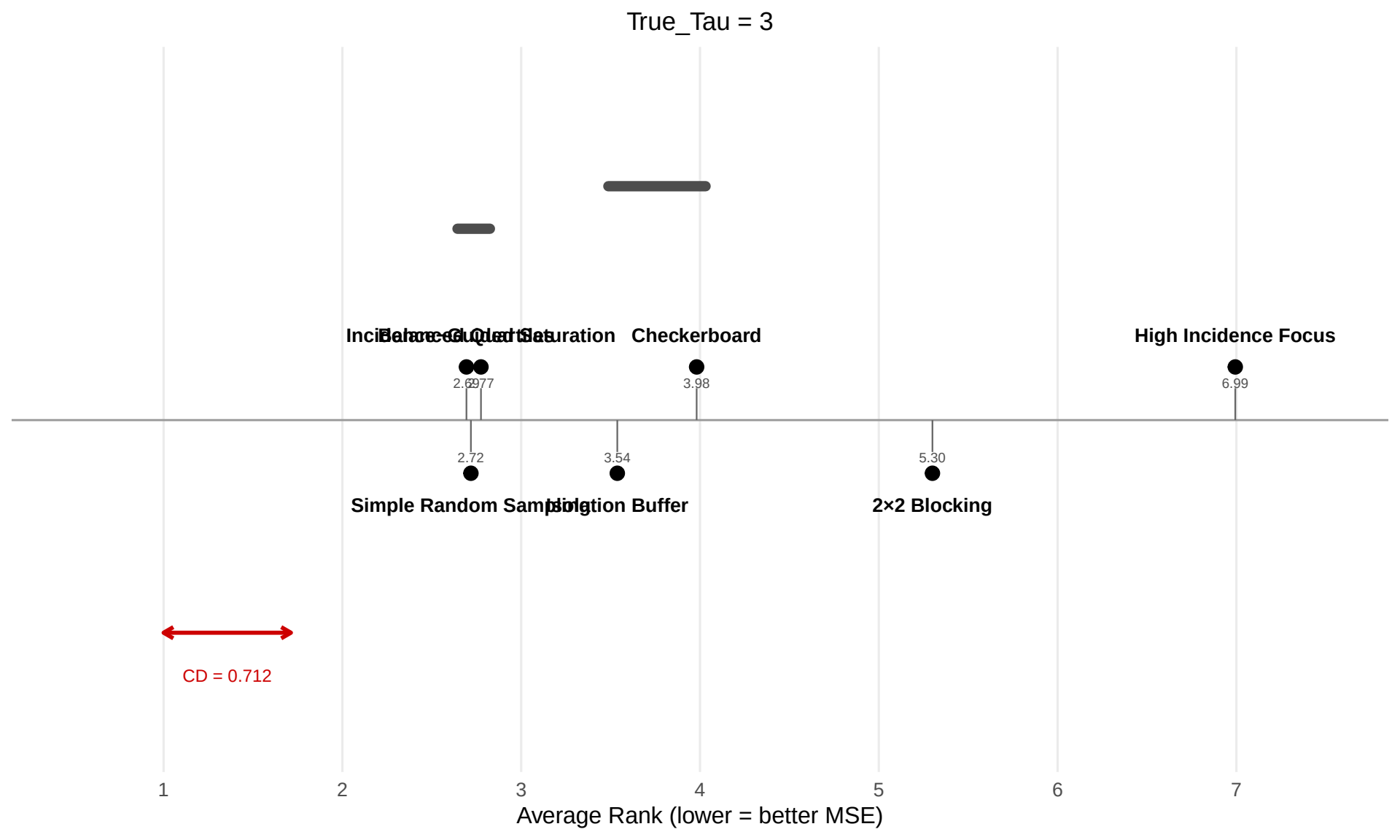

Pairwise Nemenyi p−values
2×2 Blocking
Balanced Quartiles
Checkerboard
High Incidence Focus
Incidence−Guided Saturation
Isolation Buffer
Simple Random Sampling
***
ns
**
p−value
1.00
0.50
0.10
0.001


True_Tau = 0.8
Incidence-Centered Sampling
Simple Random Sampling
Checkerboard
High Incidence Focus
2.66
2.92
3.84
7.00
2.84
3.51
5.22
Balanced Quartiles
Isolation Buffer
2×2 Blocking
CD = 0.712
1
2
3
4
5
6
7
Average Rank (lower = better MSE)


True_Tau = 1
Incidence-Stratified
Simple Random
Cluster Randomized
Saturation Sampling
Checkerboard
High Incidence Focus
2.62
2.92
3.86
7.00
2.83
3.52
5.26
Balanced Quartiles
Isolation Buffer
2×2 Blocking
CD = 0.712
1
2
3
4
5
6
7
Average Rank (lower = better MSE)


True_Tau = 1.5
Incidence-Sensitive
Simple Cluster
Random Saturation
Random Sampling
Checkerboard
High Incidence Focus
2.62
2.88
3.84
6.99
2.85
3.52
5.30
Balanced Quartiles
Isolation Buffer
2×2 Blocking
CD = 0.712
1
2
3
4
5
6
7
Average Rank (lower = better MSE)


True_Tau = 2
Incidence Gradient
Simple Random Sampling
Checkerboard
High Incidence Focus
2.65
2.83
3.92
6.99
2.79
3.52
5.29
Balanced Quartiles
Isolation Buffer
2×2 Blocking
CD = 0.712
1
2
3
4
5
6
7
Average Rank (lower = better MSE)


True_Tau = 3
Checkerboard
High Incidence Focus
3.98
6.99
2.72
3.54
5.30
Simple Random Sampling
Isolation Buffer
2×2 Blocking
CD = 0.712
1
2
3
4
5
6
7
Average Rank (lower = better MSE)